

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

```
df = pd.read_csv("basic_cyber_dataset.csv")
```

```
#Name: Kunal Dhongade
#UID: 23015035
```

```
print("\nFirst 5 Rows.")
print(df.head)
```

First 5 Rows.

```
<bound method NDFrame.head of      Unnamed: 0  connection_id  duration_ms  src_bytes  dst_bytes  \
0              0              1          120         10           8
1              1              2          350         45          30
2              2              3          800        150         120
3              3              4           60          5           4
4              4              5         1200        300        260
```

```
      failed_logins  packet_count  error_count  suspicious_score  is_encrypted  \
0              0              2           0           0.02           1
1              1              5           2           0.15           1
2              4             20          15           0.85           0
3              0              1           0           0.01           1
4              8             35          22           0.92           0
```

```
      label
0  normal
1  normal
2  attack
3  normal
4  attack >
```

```
print("\ndataset Size")
print(df.shape)
```

```
dataset Size
(5, 11)
```

```
print("\nColumn Names")
print(df.columns)
```

Column Names

```
Index(['Unnamed: 0', 'connection_id', 'duration_ms', 'src_bytes', 'dst_bytes',
      'failed_logins', 'packet_count', 'error_count', 'suspicious_score',
      'is_encrypted', 'label'],
      dtype='object')
```

```
print("\nDataset Info")
print(df.info)
```

Dataset Info

```
<bound method DataFrame.info of      Unnamed: 0  connection_id  duration_ms  src_bytes  dst_bytes  \
0              0              1          120         10           8
1              1              2          350         45          30
2              2              3          800        150         120
3              3              4           60          5           4
```

```

4          4          5          1200          300          260

      failed_logins  packet_count  error_count  suspicious_score  is_encrypted  \
0              0           2           0           0.02           1
1              1           5           2           0.15           1
2              4          20          15           0.85           0
3              0           1           0           0.01           1
4              8          35          22           0.92           0

      label
0  normal
1  normal
2  attack
3  normal
4  attack >

```

```

print("\nstatistical Summary")
print(df.describe)

```

```

statistical Summary
<bound method NDFrame.describe of      Unnamed: 0  connection_id  duration_ms  src_bytes  dst_bytes  \
0              0              1           120           10           8
1              1              2          350           45          30
2              2              3          800          150          120
3              3              4           60           5           4
4              4              5          1200          300          260

      failed_logins  packet_count  error_count  suspicious_score  is_encrypted  \
0              0           2           0           0.02           1
1              1           5           2           0.15           1
2              4          20          15           0.85           0
3              0           1           0           0.01           1
4              8          35          22           0.92           0

      label
0  normal
1  normal
2  attack
3  normal
4  attack >

```

```

print("\nMissing Values")
print(df.isnull().sum())

```

```

Missing Values
Unnamed: 0      0
connection_id    0
duration_ms      0
src_bytes        0
dst_bytes        0
failed_logins    0
packet_count     0
error_count      0
suspicious_score 0
is_encrypted     0
label            0
dtype: int64

```

```

print("\nClass Distributaion")
print(df["label"].value_counts)

```

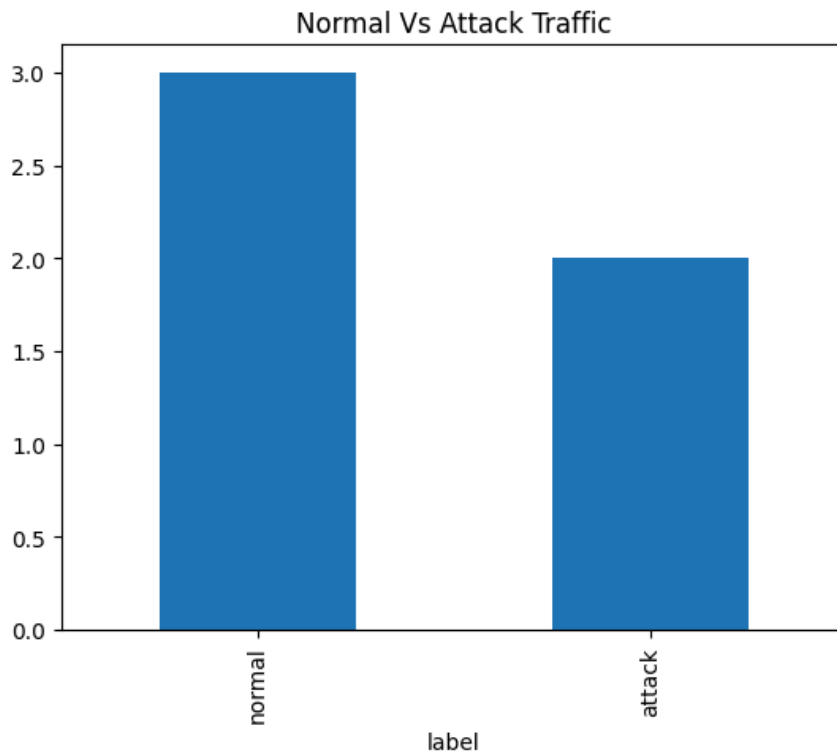
```

Class Distributaion
<bound method IndexOpsMixin.value_counts of 0    normal
1    normal
2    attack
3    normal
4    attack
Name: label, dtype: object>

```

```
df["label"].value_counts().plot(kind="bar")
plt.title("Normal Vs Attack Traffic")
```

```
Text(0.5, 1.0, 'Normal Vs Attack Traffic')
```



```
print("\n")
duration_array = np.array(df["duration_ms"])
print("\nDuration values", duration_array)
print("\nMean duration", np.mean(duration_array))
print("\nmaximum duration", np.max(duration_array))
print("\nMinimum duration", np.min(duration_array))
```

```
Duration values [ 120  350  800  60 1200]
```

```
Mean duration 506.0
```

```
maximum duration 1200
```

```
Minimum duration 60
```

```
X = df.drop(["connection_id", "label"], axis=1)
Y = df["label"]

X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.3, random_state=42
)

model = DecisionTreeClassifier()
model.fit(X_train, Y_train)
Y_pred = model.predict(X_test)
print("\n Predict Values",)
print(Y_pred)

print("\n Actual Values")
print(Y_test.values)
```

```
accuracy = accuracy_score(Y_test, Y_pred)
print("\n Accuracy", accuracy)
```

```
Predict Values
['normal' 'attack']
```

```
Actual Values
['normal' 'attack']
```

```
Accuracy 1.0
```